

HAT-P-37b

R-2639

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_160627 · created 2026-08-29T01:09:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457566.7805 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1706 +/- 0.0087
Transit depth (Rp/Rs)^2	2.91 +/- 0.3 [%]
Orbital Inclination (inc)	86.77 +/- 1.2
Airmass coefficient 1 (a1)	0.923 +/- 0.023
Airmass coefficient 2 (a2)	0.068 +/- 0.046
Scatter in the residuals of the lightcurve fit is	2.46 %
Best Comparison Star	#2 - [281, 205]
Optimal Aperture	3.28
Optimal Annulus	11.58
Transit Duration (day)	0.0968 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1706 $\pm$ 0.0087 vs 0.1394 (deviation 3.59 σ)
Duration fit vs published	0.0968 d vs 0.0983 d (deviation 0.18 σ)
Mid-time O-C	-4.8 min
Depth SNR	19.6
Residual scatter	2.46 %

Light curve

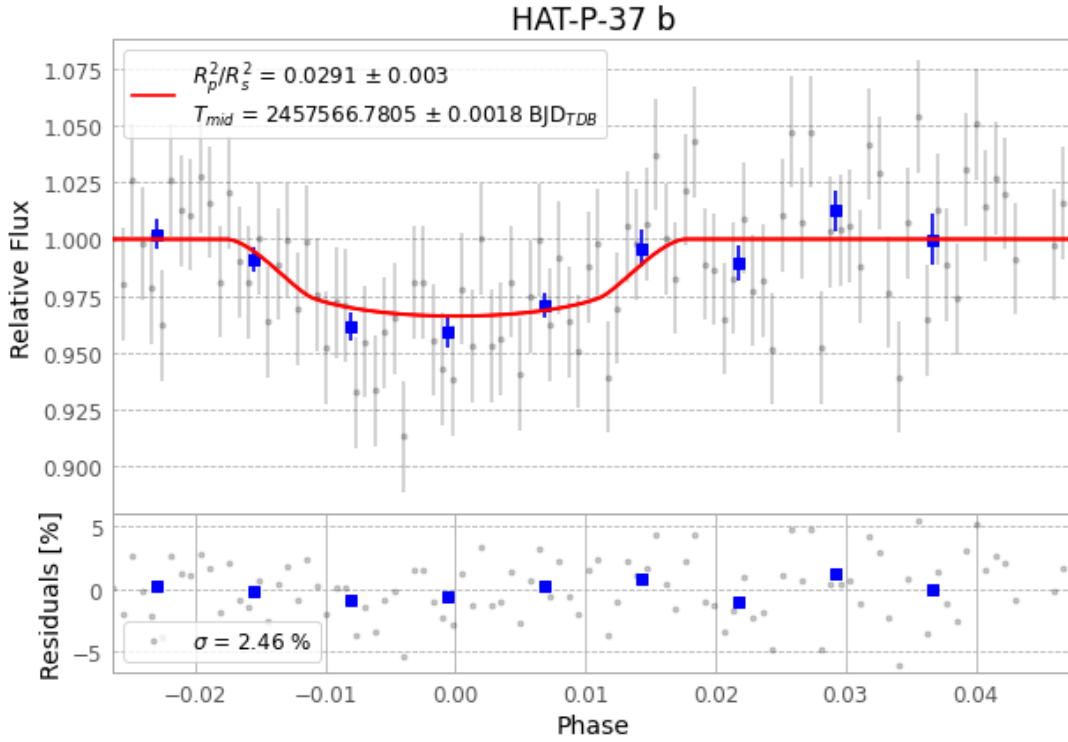


Figure 1 — FinalLightCurve_HAT-P-37 b_2016-06-26.png (SHA-256 e0630fa5f81d12a7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [167, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1c2c97dbc4db233a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

100 FITS frames (66 MB), HATP-37160627045412.FITS → HATP-37160627095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-160627.log (5d93d9e29440...), FinalLightCurve_HAT-P-37 b_2016-06-26.png (e0630fa5f81d...), FinalLightCurve_HAT-P-37 b_2016-06-26.csv (fcf0a96aaf3b...)

Compute resources

Wall time 118.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 01:09Z.